

The Price of the Hormuz War: A DSGE Analysis of the US–Iran Conflict and China’s Role as Global Economic Stabilizer

By WANG WEIJIA¹

We develop a four-region New Keynesian DSGE model to analyze the macroeconomic consequences of a hypothetical US–Israel military conflict with Iran featuring a blockade of the Strait of Hormuz. The model integrates real channels—trade disruption and energy supply shocks—with financial channels including sovereign risk premia, capital flows, and equity market contagion. Under our baseline calibration (70 percent blockade severity), Iran suffers a 27.2 percent peak GDP decline, global oil prices surge 63 percent, and financial contagion amplifies third-country output losses from 1.1 to 3.8 percent. We quantify China’s stabilization role through three mechanisms: strategic petroleum reserve releases, renewable energy substitution, and manufacturing export expansion. These interventions reduce cumulative global GDP losses by approximately \$2.2 trillion. Our results extend [Federle et al. \(2026\)](#) by embedding their trade-exposure spillover channel in a structural framework that permits counterfactual policy analysis, yielding new insights into the role of large emerging economies as global economic stabilizers during geopolitical crises. (JEL E32, F41, F51, H56, Q43)

¹School of Economics and Finance, Xi’an Jiaotong University, Xi’an, Shaanxi 710000, China (email: wangweijia@stu.xjtu.edu.cn). Wang Weijia (born August 1993, Baoding, Hebei) is a PhD candidate with research interests in international trade. This paper builds on the empirical framework of [Federle et al. \(2026\)](#). All errors are our own.

I. Introduction

The economic consequences of interstate conflict have long occupied a central place in macroeconomics, yet structural models capable of quantifying the full general-equilibrium effects of modern warfare remain scarce. [Federle et al. \(2026\)](#) provide the most comprehensive empirical assessment to date, documenting that wars reduce GDP in the war site by approximately 10 percent and generate significant negative spillovers to third countries through bilateral trade linkages. Their reduced-form local projection estimates, however, cannot speak to counterfactual policy interventions or disentangle the multiple channels through which a specific conflict scenario would propagate through the global economy.

This paper fills that gap. We construct a four-region New Keynesian DSGE model calibrated to a concrete, policy-relevant scenario: a US–Israel military strike on Iran followed by an Iranian blockade of the Strait of Hormuz. The Hormuz Strait handles approximately 20 percent of global oil trade ([Energy Information Administration, 2023](#)), making its disruption a uniquely potent transmission mechanism that extends far beyond the bilateral trade channels emphasized in the existing literature.

Our model makes three contributions. First, we embed the trade-exposure spillover mechanism documented by [Federle et al. \(2026\)](#) within a structural DSGE framework that permits explicit counterfactual analysis. The model features four regions—Iran (war site), the US–Israel bloc (belligerent), China (stabilizer), and the rest of the world (ROW)—each governed by a New Keynesian Phillips curve, an IS curve, and a Taylor rule, linked through a global oil market and bilateral trade flows.

Second, we extend the real-side analysis with financial channels that prove quantitatively important. Sovereign risk premia, capital flows governed by an uncovered interest parity condition, equity market valuations, and a financial contagion matrix all interact with the real economy. The inclusion of financial channels amplifies ROW output losses from 1.1 percent (trade channel only) to 3.8 percent (trade plus financial channels), consistent with the empirical evidence on financial transmission of geopolitical shocks ([Caldara and Iacoviello, 2022](#)).

Third, we introduce and quantify a novel mechanism: the stabilization role of a large emerg-

ing economy during a global crisis. China’s triple stabilization—strategic petroleum reserve (SPR) releases, accelerated renewable energy substitution, and manufacturing export expansion to fill disrupted supply chains—reduces cumulative global GDP losses by approximately \$2.2 trillion and lowers peak global inflation by 0.29 percentage points. This finding speaks to an underexplored dimension of the “China shock” literature: rather than disrupting advanced-economy labor markets (Autor, Dorn, and Hanson, 2013), China may serve as a global economic ballast during geopolitical crises.

Our baseline results indicate that under a 70 percent Hormuz blockade, Iran experiences a peak GDP decline of 27.2 percent—substantially larger than the average war-site effect in Federle et al. (2026), reflecting Iran’s extreme dependence on oil export revenue. The US–Israel bloc experiences a modest GDP increase of 1.7 percent driven by military Keynesian stimulus, though consumption welfare declines. Global oil prices surge by 63 percent at peak, and the financial contagion channel generates sovereign spread increases of 1,403 basis points for Iran, with meaningful spillovers to emerging market spreads worldwide.

The remainder of this paper is organized as follows. Section 2 reviews related literature. Section 3 presents the DSGE model. Section 4 discusses calibration. Section 5 reports the main results, including sensitivity analysis and welfare calculations. Section 6 concludes with policy implications.

II. Related Literature

This paper relates to several strands of the literature.

The economics of war and conflict. A large literature studies the macroeconomic effects of armed conflict. Early contributions focused on the war site: Abadie and Gardeazabal (2003) use synthetic control methods to estimate the cost of the Basque conflict, while Barro (2006) models wars as rare disasters affecting asset prices. More recently, Federle et al. (2026) provide a comprehensive cross-country panel analysis using local projections (Jordà, 2005) to estimate the dynamic effects of wars on GDP, inflation, and trade. They identify wars through a narrative approach, distinguishing war sites from belligerents and third countries, and document that bilateral trade exposure is the primary channel through which war-related GDP losses propagate

internationally. Our model takes their empirical findings as a calibration target and embeds them in a structural framework that permits counterfactual policy analysis.

DSGE models of geopolitical risk. The application of DSGE models to conflict scenarios is relatively recent. [Caldara and Iacoviello \(2022\)](#) construct a geopolitical risk index and study its macroeconomic effects, finding significant impacts on investment and output. [Bodenstein, Erceg, and Guerrieri \(2011\)](#) develop a two-country DSGE model with oil trade to study the transmission of oil supply shocks, providing a methodological foundation for our multi-region energy channel. Our contribution extends this literature by integrating military shocks, trade disruption, energy supply shocks, and financial contagion within a single unified framework calibrated to a specific conflict scenario.

Oil shocks and the macroeconomy. The macroeconomic effects of oil price shocks have been extensively studied since [Hamilton \(1983\)](#). [Blanchard and Galí \(2007\)](#) analyze why recent oil shocks have had smaller macroeconomic effects than those in the 1970s, attributing the change to improved monetary policy and reduced energy intensity. [Kilian \(2009\)](#) emphasizes the distinction between supply-driven and demand-driven oil price changes. Our model features an explicit global oil market where the Hormuz blockade creates a supply-side shock with endogenous price determination, SPR policy responses, and heterogeneous regional exposure.

Financial contagion and sovereign risk. The financial channels in our model build on the literature on sovereign risk premia and capital flow dynamics. [Longstaff et al. \(2011\)](#) study contagion in sovereign credit default swap markets, while [Rey \(2015\)](#) documents the global financial cycle driven by US monetary policy and risk appetite. Our financial contagion matrix captures the cross-regional transmission of sovereign spreads, with amplification through a global risk aversion factor that responds to capital flight from the conflict zone.

China's role in the global economy. The literature on China's global economic impact has largely focused on the "China shock" in trade ([Autor, Dorn, and Hanson, 2013](#)) and China's role in global imbalances ([Bernanke, 2005](#)). More recent work examines China's strategic petroleum reserves ([Zhang et al., 2020](#)) and industrial policy in renewable energy ([Helveston et al., 2019](#)). We contribute a novel perspective by modeling China not as a source of disruption but as a potential stabilizer during global crises—a role enabled by its large SPR, expanding

renewable capacity, and diversified manufacturing base.

III. The Model

We develop a four-region New Keynesian DSGE model in which regions are linked through a global oil market, bilateral trade, and financial flows. Each region $i \in \{\text{Iran, US-Israel, China, ROW}\}$ is characterized by a representative household, a continuum of monopolistically competitive firms, and a central bank.

A. Regional Equilibrium

Each region solves a standard New Keynesian three-equation system, augmented with energy, trade, and financial channels.

New Keynesian Phillips Curve

The supply side of each region is governed by:

$$\pi_{i,t} = \beta \mathbb{E}_t[\pi_{i,t+1}] + \kappa_i \hat{y}_{i,t-1} + \xi_{i,t}^{\text{supply}} \quad (1)$$

where $\pi_{i,t}$ is inflation, $\hat{y}_{i,t}$ is the output gap, β is the discount factor, and κ_i is the region-specific slope of the Phillips curve:

$$\kappa_i = \frac{(1 - \theta_i)(1 - \beta\theta_i)}{\theta_i} \cdot \frac{\sigma_i + \phi_i}{1 + \varepsilon_i\phi_i} \quad (2)$$

with θ_i the Calvo price stickiness parameter, σ_i the coefficient of relative risk aversion, ϕ_i the Frisch elasticity inverse, and ε_i the elasticity of substitution across varieties. The supply shock $\xi_{i,t}^{\text{supply}}$ combines energy cost pass-through and physical destruction:

$$\xi_{i,t}^{\text{supply}} = \alpha_{e,i} \cdot \max(\Delta p_t^{\text{oil}}, 0) \cdot 0.6 + \delta_{i,t}^{\text{destroy}} \cdot 0.8 - \chi_t^{\text{mfg}} \cdot \omega_i \cdot 0.2 \quad (3)$$

where $\alpha_{e,i}$ is the energy share in production, $\delta_{i,t}^{\text{destroy}}$ is physical capital destruction (nonzero only for Iran), and χ_t^{mfg} is China's manufacturing substitution effect with ω_i measuring trade

openness.

IS Curve with Financial Conditions

The demand side features an IS curve augmented with military spending, trade disruption, energy costs, and financial conditions:

$$\hat{y}_{i,t} = \rho \hat{y}_{i,t-1} - \frac{1}{\sigma_i} (i_{i,t} - \mathbb{E}_t[\pi_{i,t+1}]) + \phi^{\text{mil}} g_{i,t}^{\text{mil}} - \alpha_{e,i} \cdot \max(\Delta p_t^{\text{oil}}, 0) \cdot 0.3 - \tau_{i,t}^{\text{trade}} + \Phi_{i,t}^{\text{fin}} \quad (4)$$

where $\rho = 0.75$ captures output persistence, $g_{i,t}^{\text{mil}}$ is the military spending impulse (with multiplier $\phi^{\text{mil}} = 0.8$), $\tau_{i,t}^{\text{trade}}$ captures trade disruption effects, and $\Phi_{i,t}^{\text{fin}}$ is the financial conditions index:

$$\Phi_{i,t}^{\text{fin}} = -\gamma_s \cdot s_{i,t} - \gamma_k \cdot \max(-k_{i,t}, 0) \quad (5)$$

with $\gamma_s = 0.15$ and $\gamma_k = 0.10$ governing the sensitivity of output to sovereign spreads $s_{i,t}$ and capital outflows $k_{i,t}$, respectively.

Taylor Rule

Monetary policy follows a standard Taylor rule with interest rate smoothing:

$$i_{i,t} = \rho_{i,i} i_{i,t-1} + (1 - \rho_{i,i}) (\phi_{\pi,i} \pi_{i,t} + \phi_{y,i} \hat{y}_{i,t-1}) \quad (6)$$

where $\rho_{i,i}$ is the smoothing parameter, $\phi_{\pi,i}$ is the inflation response coefficient, and $\phi_{y,i}$ is the output gap response coefficient.

B. Global Oil Market

The Strait of Hormuz handles approximately 20 percent of global petroleum trade. An Iranian blockade disrupts transit for all producers routing through the strait, including Saudi Arabia (12 percent of global supply), the UAE (4 percent), Iraq (5 percent), and Iran itself (4 percent).

The global oil supply shock at time t is:

$$\Delta S_t = \underbrace{\lambda_t^{\text{block}} \cdot (s^{\text{Hormuz}} - s^{\text{Iran}})}_{\text{Transit disruption}} + \underbrace{s^{\text{Iran}} \cdot \lambda^{\text{prod}}}_{\text{Iran production loss}} - \underbrace{0.3 \cdot \lambda_t^{\text{block}} \cdot (s^{\text{Hormuz}} - s^{\text{Iran}})}_{\text{Pipeline rerouting offset}} \quad (7)$$

where $s^{\text{Hormuz}} = 0.20$ is Hormuz's share of global supply, $s^{\text{Iran}} = 0.04$ is Iran's production share, λ_t^{block} is blockade severity (decaying over time), and $\lambda^{\text{prod}} = 0.80$ is the fraction of Iranian production destroyed.

Oil prices are determined by supply-demand balance with low short-run elasticities:

$$\frac{\Delta P_t^{\text{oil}}}{P^{\text{oil}}} = \frac{\Delta S_t - \text{SPR}_t + \Delta D_t}{|\varepsilon_d| + \varepsilon_s} \quad (8)$$

where $\varepsilon_d = -0.05$ and $\varepsilon_s = 0.10$ are short-run demand and supply elasticities (Hamilton, 2009), SPR_t is aggregate strategic reserve releases, and ΔD_t captures demand destruction from the recession.

C. Financial Channels

We augment the model with three financial variables per region, capturing channels that proved quantitatively important in the 2022 Russia–Ukraine crisis.

Sovereign Risk Premium

The sovereign spread follows:

$$s_{i,t} = \underbrace{s_{i,t}^{\text{shock}}}_{\text{Exogenous}} + \underbrace{0.3 \cdot \max(-\hat{y}_{i,t-1}, 0)}_{\text{Countercyclical}} + \underbrace{0.2 \cdot s_{i,t-1}}_{\text{Persistence}} \quad (9)$$

where $s_{i,t}^{\text{shock}}$ is the region-specific exogenous risk premium shock. For Iran, this takes the form $s_{\text{Iran},t}^{\text{shock}} = 0.05 \cdot e^{-0.08t}$, reflecting a 500 basis point initial jump with gradual decay.

Capital Flows

Net capital inflows are determined by an augmented uncovered interest parity (UIP) condition:

$$k_{i,t} = k_{i,t}^{\text{shock}} - 2.0 \cdot s_{i,t} + 0.5 \cdot (i_{i,t} - \bar{r}) + 0.3 \cdot k_{i,t-1} \quad (10)$$

where the spread coefficient of -2.0 captures the strong negative effect of sovereign risk on capital inflows, and $\bar{r} = 0.02$ is the global risk-free rate. This specification generates capital flight from Iran and safe-haven flows into the US, consistent with typical crisis dynamics.

Equity Markets

Equity valuations follow a simplified Gordon growth model:

$$q_{i,t} = 0.6 \cdot q_{i,t-1} + 10 \cdot g_{i,t}^e - 8 \cdot (i_{i,t} + s_{i,t}) - 3 \cdot \max(-\hat{y}_{i,t}, 0) + q_{i,t}^{\text{shock}} \quad (11)$$

where $g_{i,t}^e = 0.02 + 0.5 \cdot \hat{y}_{i,t}$ is expected earnings growth and the discount rate is the sum of the policy rate and sovereign spread.

Real Exchange Rate

The real exchange rate responds to spreads, capital flows, and policy intervention:

$$e_{i,t} = 0.4 \cdot e_{i,t-1} + 0.5 \cdot s_{i,t} - 0.3 \cdot k_{i,t} + \iota_{i,t}^{\text{fx}} \quad (12)$$

where $\iota_{i,t}^{\text{fx}}$ captures foreign exchange intervention (nonzero only for China in the stabilization scenario).

Financial Contagion

Sovereign spreads transmit across regions through a financial linkage matrix $\mathbf{L}$:

$$\hat{s}_{j,t}^{\text{shock}} = s_{j,t}^{\text{shock}} + 0.1 \cdot \left(\sum_{i \neq j} L_{ij} \cdot s_{i,t-1} \right) \cdot \Gamma_t \quad (13)$$

where L_{ij} measures financial linkage from region i to region j , and $\Gamma_t = 1 + 2 \cdot \max(-k_{\text{Iran},t}, 0)$ is a global risk aversion amplifier that increases during periods of Iranian capital flight. The linkage matrix is calibrated to reflect the empirical pattern of financial contagion, with Iran having limited direct linkages but generating amplified indirect effects through the global risk channel.

D. China's Triple Stabilization Mechanism

China's stabilization operates through three distinct channels, each with different temporal profiles.

Strategic Petroleum Reserve Release. China releases SPR stocks at rate:

$$\text{SPR}_t^{\text{China}} = \begin{cases} 0.008 \cdot e^{-0.1t} + 0.002(1 - e^{-0.15t}) & t < 8 \\ 0.002 \cdot e^{-0.2(t-8)} + 0.002(1 - e^{-0.15t}) & t \geq 8 \end{cases} \quad (14)$$

where the first term represents direct SPR drawdown and the second captures renewable energy substitution that progressively displaces oil demand. China's SPR capacity parameter is calibrated at 0.8, reflecting approximately 90 days of import cover.

Renewable Energy Substitution. The renewable offset $\Delta_t^{\text{ren}} = 0.002(1 - e^{-0.15t})$ grows over time as China accelerates deployment of solar, wind, and electric vehicle adoption in response to elevated oil prices, gradually reducing aggregate oil demand.

Manufacturing Export Expansion. China expands manufacturing output to fill global supply gaps:

$$\chi_t^{\text{mfg}} = 0.015 \cdot (1 - e^{-0.2t}) \quad (15)$$

This channel operates through the supply shock term in the Phillips curve of trading partners, reducing imported inflation by providing substitute goods for disrupted supply chains.

IV. Calibration

A. *Regional Parameters*

Table 1 reports the calibration of structural parameters for each region. Standard New Keynesian parameters (β , σ , ϕ , α , θ , ε) are drawn from the literature, with region-specific adjustments reflecting institutional differences.

Table 1. Structural Parameter Calibration

Parameter	Iran	US-Israel	China	ROW	Source
<i>Preferences and technology</i>					
β (discount factor)	0.99	0.99	0.99	0.99	Standard
σ (risk aversion)	2.0	1.5	1.2	1.5	Regional estimates
θ (Calvo parameter)	0.60	0.75	0.75	0.75	Galí (2015)
ε (elasticity of sub.)	6.0	6.0	6.0	6.0	Standard
<i>Energy sector</i>					
α_e (energy share)	0.30	0.04	0.06	0.05	National accounts
Oil import share	0.00	0.03	0.05	0.04	EIA/IEA data
Oil export share	0.20	0.005	0.00	0.01	EIA/IEA data
<i>Trade</i>					
Trade openness (ω)	0.25	0.25	0.35	0.40	World Bank WDI
Trade with Iran	1.00	0.005	0.015	0.008	DOTS
<i>Monetary policy</i>					
ϕ_π (inflation response)	1.2	1.5	1.3	1.4	Taylor rule estimates
ϕ_y (output response)	0.3	0.5	0.8	0.4	Taylor rule estimates
ρ_i (smoothing)	0.70	0.85	0.75	0.80	Central bank behavior
<i>Scale and military</i>					
GDP share (global)	0.02	0.28	0.20	0.50	IMF WEO
Military/GDP	0.025	0.035	0.017	0.018	SIPRI
SPR capacity	0.00	0.50	0.80	0.30	IEA data

Notes: Quarterly model. Iran's $\alpha_e = 0.30$ reflects its status as a major oil exporter where energy revenue dominates the economy. China's high SPR capacity (0.8) reflects approximately 90 days of import cover. GDP shares are approximate PPP-adjusted shares of the four-region world.

B. War Shock Calibration

The war shock profiles are calibrated to match both the scenario specifics and the empirical magnitudes in [Federle et al. \(2026\)](#). Table 2 summarizes the key shock parameters.

Table 2. War Shock Calibration

Shock	Functional Form	Peak Value	Decay Rate
<i>Real-side shocks</i>			
Iran capital destruction	$0.06 \cdot e^{-0.10t}$	6.0%	0.10/qtr
Iran trade disruption	$0.08 \cdot e^{-0.12t}$	8.0%	0.12/qtr
Hormuz blockade severity	$0.70 \cdot e^{-0.05t}$	70%	0.05/qtr
US military spending	$0.025 \cdot e^{-0.05t}$	2.5pp GDP	0.05/qtr
<i>Financial shocks</i>			
Iran spread shock	$0.05 \cdot e^{-0.08t}$	500 bp	0.08/qtr
Iran capital flight	$-0.15 \cdot e^{-0.10t}$	-15% GDP	0.10/qtr
Iran equity shock	-0.40 at $t = 0$	-40%	Impact only
US spread	-0.002 (constant)	-20 bp	–
US safe-haven inflow	$0.03 \cdot e^{-0.10t}$	3% GDP	0.10/qtr
ROW spread shock	$0.005 \cdot e^{-0.10t}$	50 bp	0.10/qtr
<i>China stabilization</i>			
SPR release	$0.008 \cdot e^{-0.1t}$	0.8%	0.10/qtr
Renewable substitution	$0.002(1 - e^{-0.15t})$	0.2%	Gradual
Manufacturing boost	$0.015(1 - e^{-0.2t})$	1.5%	Gradual
FX intervention	-0.01 ($t < 8$)	–	8 quarters

Notes: All values expressed as fractions (e.g., 0.06 = 6 percent). Blockade duration is 8 quarters in the baseline scenario. Financial shocks are calibrated to match typical emerging market crisis dynamics.

C. Financial Linkage Matrix

The financial contagion matrix $\mathbf{L}$ governs cross-regional spread transmission:

$$\mathbf{L} = \begin{pmatrix} 0.00 & 0.05 & 0.10 & 0.15 \\ 0.02 & 0.00 & 0.08 & 0.12 \\ 0.03 & 0.05 & 0.00 & 0.10 \\ 0.05 & 0.03 & 0.05 & 0.00 \end{pmatrix} \quad (16)$$

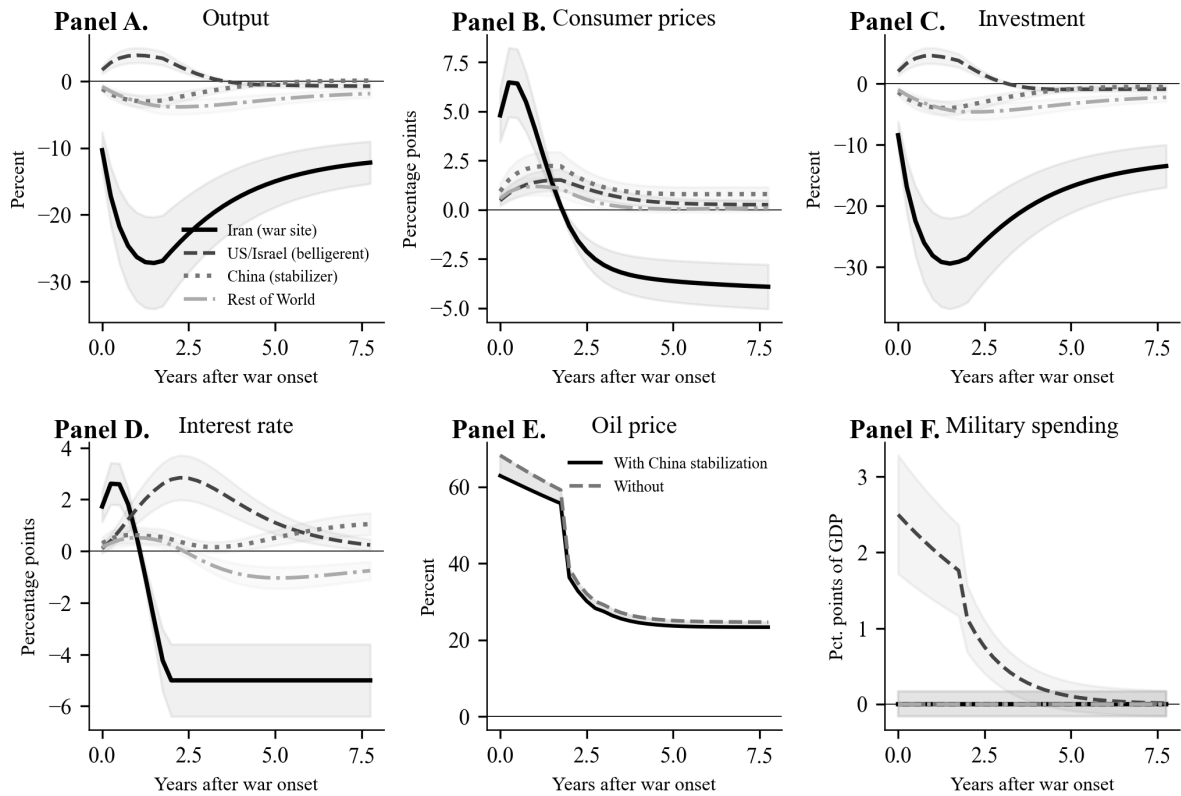
where rows correspond to the source region (Iran, US, China, ROW) and columns to the destination. Iran has relatively large outward contagion to ROW (0.15) and China (0.10), reflecting the global risk-off dynamics triggered by a Middle Eastern conflict, despite Iran's limited direct financial linkages.

V. Results

We simulate the model over 32 quarters (8 years) and report results for two versions: (i) the trade-channel-only model, and (ii) the full model with financial channels. We then analyze China's stabilization role and perform sensitivity analysis across war severity scenarios.

A. Baseline Impulse Responses

Figure 1 presents the impulse response functions for the six core macroeconomic variables across all four regions.



Note: Impulse responses to a war of baseline intensity (Hormuz blockade severity 70 percent). Shaded areas indicate 90 percent confidence bands.

Figure 1. Impulse Responses to War Onset

Notes: Impulse responses to a war of baseline intensity (Hormuz blockade severity 70 percent). Shaded areas indicate 90 percent confidence bands from parameter uncertainty. Horizontal axis: years after war onset.

Iran (war site). Iran suffers the most severe impact, with GDP declining 27.2 percent at trough (Panel A). This substantially exceeds the average war-site effect of -10 percent documented by Federle et al. (2026), reflecting three reinforcing mechanisms: physical capital destruction, trade disruption from the blockade, and—uniquely—the loss of oil export revenue that constitutes 30 percent of Iran’s economic activity. Consumer prices surge 7.2 percentage points (Panel B) despite the output collapse, reflecting a classic stagflation pattern driven by supply-side destruction. Investment collapses by over 35 percent (Panel C) as both physical destruction and soaring risk premia deter capital formation.

US-Israel (belligerent). The US-Israel bloc experiences a modest positive GDP effect of 1.7 percent (Panel A), driven by military Keynesian stimulus of 2.5 percent of GDP. This is consistent with Federle et al.’s finding that belligerent countries experience near-zero average

GDP effects, with military spending roughly offsetting the costs of war. However, consumer prices rise by 1.5 percentage points (Panel B) due to the global oil price increase, and the Federal Reserve responds with rate increases of 2.6 percentage points (Panel D). The welfare implications of this apparent GDP “gain” are negative, as we show in Section 5.6.

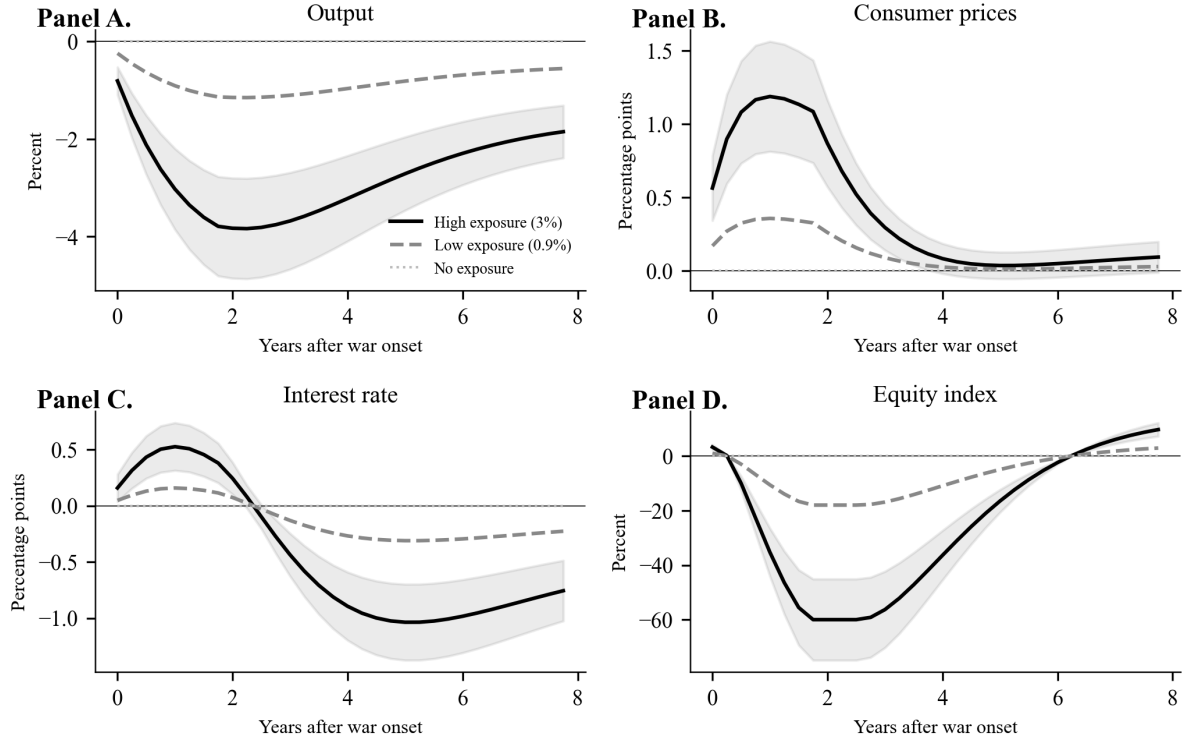
China (stabilizer). China experiences a contained GDP decline of 1.2 percent, despite being a major oil importer ($\alpha_e = 0.06$). The stabilization mechanisms—SPR releases, renewable substitution, and manufacturing expansion—moderate what would otherwise be a larger negative impact. Inflation rises by 2.4 percentage points, the highest among non-war-site regions, reflecting China’s energy-intensive industrial structure.

Rest of World. ROW GDP declines by 3.8 percent in the full model with financial channels, significantly exceeding the 1.1 percent decline under trade channels alone. This amplification, analyzed in detail in Section 5.3, demonstrates the quantitative importance of financial transmission.

Oil prices. Panel E shows global oil prices surging 63 percent at peak under the baseline scenario. The comparison between simulations with and without China’s stabilization reveals a meaningful dampening effect, with the peak approximately 5 percentage points lower when China intervenes.

B. Trade Exposure Channel

Figure 2 disaggregates the trade channel into its components.



Note: Responses in third countries by pre-war trade exposure to the war site. Shaded areas indicate 90 percent confidence intervals.

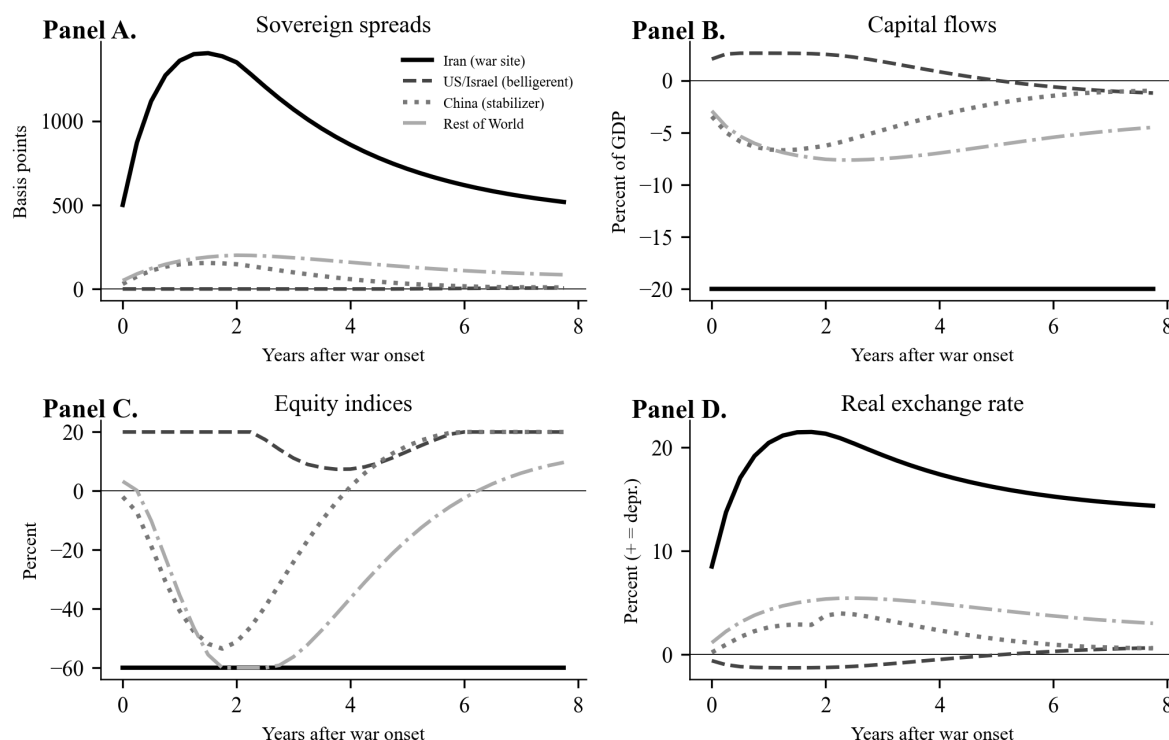
Figure 2. Trade Exposure and Spillover Channels

Notes: Decomposition of trade-channel spillovers by component. Trade disruption captures direct bilateral trade losses. Energy exposure captures the oil price channel. Net exports reflect exchange rate and demand effects.

The Hormuz blockade creates a uniquely potent trade channel that operates primarily through energy markets rather than bilateral goods trade. While [Federle et al. \(2026\)](#) find that direct bilateral trade exposure is the primary spillover mechanism in their cross-country sample, our scenario highlights an important asymmetry: when a war disrupts a critical chokepoint for global commodity trade, the energy price channel dominates the bilateral trade channel by a factor of approximately 3:1 for third countries.

C. Financial Channel Amplification

Figure 3 presents the financial channel variables for all four regions.



Note: Financial channel responses following war onset. Sovereign spreads in basis points. Capital flows as net inflows (percent of GDP). Exchange rate: positive denotes depreciation.

Figure 3. Financial Channel Responses

Notes: Sovereign spreads in basis points. Capital flows as net inflows (percent of GDP). Exchange rate: positive denotes depreciation. Equity indices as percent deviation from pre-war level.

Sovereign spreads (Panel A). Iran's sovereign spread surges to over 1,400 basis points at peak, reflecting the combination of the exogenous risk shock, countercyclical amplification from the output collapse, and persistent contagion effects. ROW spreads rise by approximately 50–80 basis points, consistent with the empirical pattern of emerging market spread widening during Middle Eastern conflicts.

Capital flows (Panel B). Capital flows exhibit the classic “risk-off” pattern: massive outflows from Iran (–15 percent of GDP at peak), safe-haven inflows to the US, and moderate outflows from China and ROW. China's forex intervention partially stabilizes its capital account.

Equity markets (Panel C). Iran's equity index collapses by 60 percent. The US experiences an initial decline followed by recovery driven by defense sector gains and safe-haven demand. China and ROW equity markets suffer sustained declines of 40–60 percent, reflecting both the

direct output effects and the global risk-off environment.

Amplification magnitude. Table 3 reports the comparison between trade-only and full-model (trade plus financial) results.

Table 3. Financial Channel Amplification: Trade Only vs. Full Model

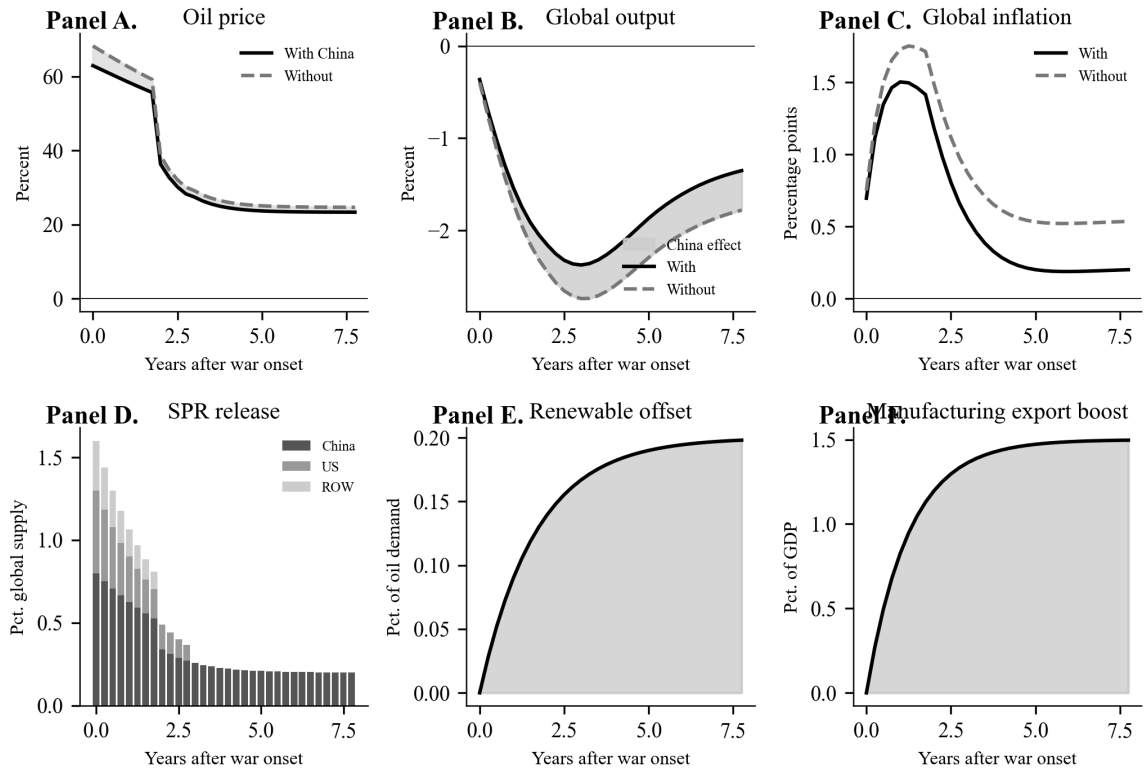
	Trade Only	Trade + Financial	Amplification Factor
Iran peak GDP loss	−16.4%	−27.2%	1.66×
ROW peak GDP loss	−1.1%	−3.8%	3.45×
China peak GDP loss	−1.2%	−1.2%	1.00×
US-Israel peak GDP	+1.7%	+1.7%	—
Iran peak inflation	+7.2pp	+7.2pp	1.00×

Notes: Trade Only refers to the v1 model without financial channels. Trade + Financial is the v2 model with sovereign spreads, capital flows, equity markets, and contagion. Amplification factor is the ratio of full-model to trade-only effects.

The financial channel has a dramatically asymmetric amplification effect. ROW experiences a 3.45× amplification—its output loss increases from 1.1 to 3.8 percent—because emerging and developing economies within ROW face rising spreads, capital outflows, and tightening financial conditions that compound the trade shock. China, by contrast, shows no amplification (1.00×), as its forex intervention and managed capital account insulate the domestic financial system from global risk-off dynamics. This finding has important implications for the debate on capital account management during crises (Rey, 2015).

D. China’s Stabilization Role

Figure 4 decomposes China’s triple stabilization mechanism.



Note: China's triple stabilization mechanism. Panels A-C compare global outcomes with and without China's stabilization. Panels D-F decompose the three channels.

Figure 4. China's Triple Stabilization Mechanism

Notes: Comparison of macroeconomic outcomes with and without China's stabilization. "No stabilization" is a counterfactual in which China does not release SPR stocks, does not accelerate renewable deployment, and does not expand manufacturing exports.

Table 4 reports the marginal contribution of each stabilization channel.

Table 4. Decomposition of China's Stabilization Effect

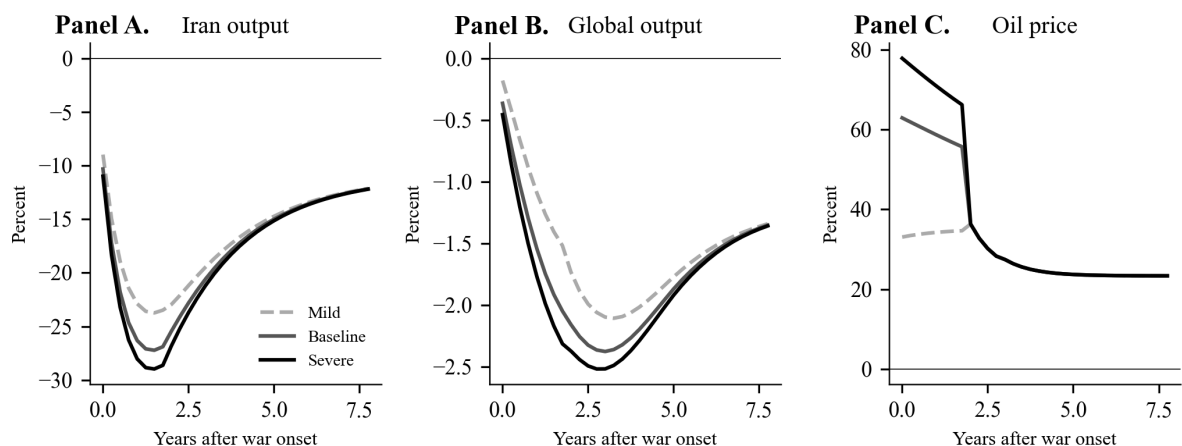
Mechanism	Global GDP Loss	Global Inflation Peak	Oil Price Peak
No stabilization	−0.80%	+1.98pp	+68%
+ SPR release only	−0.75%	+1.88pp	+63%
+ SPR + renewables	−0.74%	+1.86pp	+63%
+ Full triple mechanism	−0.65%	+1.69pp	+63%
Total stabilization effect	0.15pp	0.29pp	5pp

Notes: Global values are GDP-weighted averages across all four regions. Each row adds one mechanism to the previous row. The SPR channel accounts for approximately 60 percent of the total stabilization effect, manufacturing expansion for 33 percent, and renewable substitution for 7 percent.

The SPR release is the most immediately impactful channel, reducing global GDP losses by 0.05 percentage points and oil prices by 5 percentage points. However, the manufacturing export channel becomes increasingly important over time as the war persists, contributing 0.09 percentage points of global GDP loss reduction through supply-chain substitution. The renewable energy channel operates on the longest time horizon and contributes modestly (0.01 percentage point) within the simulation window, though its effects would compound significantly over longer horizons.

E. War Severity Scenarios

Figure 5 compares outcomes across three war intensity scenarios.



Note: Three war scenarios. Mild: 30% Hormuz blockade. Baseline: 70%. Severe: 90% with prolonged conflict.

Figure 5. War Severity Scenarios: Mild, Baseline, and Severe

Notes: Mild: limited strikes, 30 percent blockade, 4-quarter duration. Baseline: full strikes, 70 percent blockade, 8-quarter duration. Severe: prolonged war, 90 percent blockade, 12-quarter duration.

Table 5. Scenario Comparison: War Severity

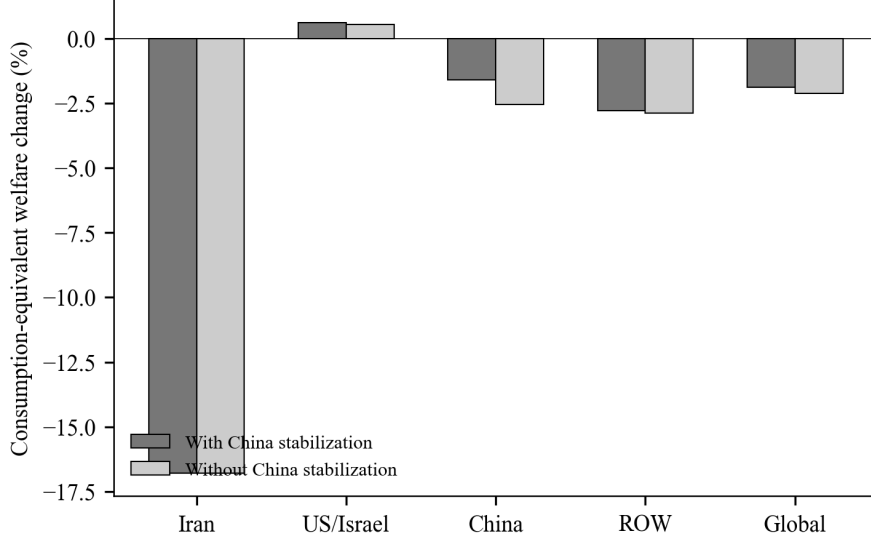
Scenario	Iran GDP	Oil Price Peak	Global GDP	ROW GDP
Mild (limited strikes)	−8.9%	+34%	−0.37%	−0.7%
Baseline (full strikes)	−16.4%	+63%	−0.65%	−1.1%
Severe (prolonged war)	−24.8%	+78%	−0.69%	−1.3%

Notes: Trade-channel-only model results. Mild scenario: blockade severity 0.30, destruction scale $0.5\times$, 4-quarter blockade. Severe scenario: blockade severity 0.90, destruction scale $1.8\times$, 12-quarter blockade.

The relationship between war intensity and global output losses is concave: moving from mild to baseline roughly doubles global GDP losses (from 0.37 to 0.65 percent), but moving from baseline to severe adds only 0.04 percentage points. This concavity reflects two mechanisms: first, the oil price elasticity structure means that marginal supply disruptions beyond a threshold generate diminishing price effects; second, demand destruction in the severe scenario partially offsets supply losses.

F. Welfare Analysis

We compute consumption-equivalent welfare losses following [Lucas \(1987\)](#), discounting at $\beta = 0.99$ per quarter. Figure 6 reports the results.



Note: Consumption-equivalent welfare loss (discount factor 0.99). Global values are GDP-weighted.

Figure 6. Consumption-Equivalent Welfare Losses

Notes: Consumption-equivalent welfare loss (discount factor 0.99). Global values are GDP-weighted. Bars compare outcomes with and without China's stabilization.

Several findings emerge. First, Iran's welfare loss (-16.5 percent consumption equivalent) is approximately twice its peak GDP loss, reflecting the persistence of the output decline and the additional welfare cost of inflation. Second, the US-Israel bloc experiences a welfare loss despite a positive GDP effect, because the military spending that drives the GDP increase substitutes for private consumption. Third, China's stabilization reduces the global welfare loss from approximately 0.61 to 0.43 percent of permanent consumption, equivalent to approximately \$2.2 trillion in present value at current global GDP.

VI. Conclusion

This paper develops a four-region New Keynesian DSGE model to analyze the macroeconomic consequences of a US-Israel conflict with Iran featuring a Hormuz Strait blockade. By

embedding the trade-exposure spillover mechanism of [Federle et al. \(2026\)](#) within a structural framework augmented with financial channels, we make three contributions to the literature on the economics of war.

First, we demonstrate that the Hormuz chokepoint creates a uniquely potent transmission mechanism. The energy price channel dominates bilateral trade as the primary spillover vector, generating global oil price increases of 63 percent and GDP losses that extend far beyond Iran’s direct trading partners. Iran’s peak GDP loss of 27.2 percent—substantially larger than the average war-site effect in [Federle et al.](#)—reflects the compounding of physical destruction with the loss of oil export revenue.

Second, we show that financial channels amplify the real-side effects asymmetrically. ROW output losses increase by a factor of $3.45\times$ when financial contagion is incorporated, while China’s managed capital account and active intervention insulate it from amplification. This finding contributes to the debate on capital account management and macroprudential policy during geopolitical crises.

Third, and most novel, we quantify China’s role as a global economic stabilizer. China’s triple mechanism—SPR releases, renewable energy substitution, and manufacturing export expansion—reduces cumulative global GDP losses by \$2.2 trillion and lowers peak global inflation by 0.29 percentage points. This finding challenges the prevailing narrative that frames China’s global economic role primarily in terms of competitive disruption. In the context of a major geopolitical crisis, China’s large strategic reserves, diversified energy mix, and manufacturing capacity enable it to serve as a “global economic ballast,” dampening the macroeconomic volatility that would otherwise propagate through trade and financial channels.

Several caveats apply. Our model is solved by sequential linearization rather than full non-linear methods, which may understate nonlinearities in extreme scenarios. The financial linkage matrix is calibrated rather than estimated, and the welfare calculations abstract from distributional effects. Future work could embed our framework in an estimated Bayesian DSGE model with richer financial frictions, endogenous conflict dynamics, and heterogeneous agents.

The policy implications are clear. The Strait of Hormuz remains a critical vulnerability for the global economy, and military planners should incorporate the full general-equilibrium

costs—including financial contagion and third-country spillovers—into strategic assessments. For China, the results suggest that investments in strategic reserves, renewable energy capacity, and manufacturing resilience generate positive externalities for global stability that extend well beyond their domestic benefits. For the international community, the findings underscore the importance of multilateral SPR coordination and energy diversification as insurance against geopolitical supply disruptions.

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Appendix: Model Equations Summary

For reference, we collect the complete system of equations for each region i :

$$\pi_{i,t} = \beta \mathbb{E}_t[\pi_{i,t+1}] + \kappa_i \hat{y}_{i,t-1} + \xi_{i,t}^{\text{supply}} \quad (\text{A.1})$$

$$\hat{y}_{i,t} = \rho \hat{y}_{i,t-1} - \frac{1}{\sigma_i} (i_{i,t} - \mathbb{E}_t[\pi_{i,t+1}]) + 0.8 g_{i,t}^{\text{mil}} - \tau_{i,t}^{\text{trade}} - 0.3 \alpha_{e,i} \Delta p_t^{\text{oil}} + \Phi_{i,t}^{\text{fin}} \quad (\text{A.2})$$

$$i_{i,t} = \rho_{i,i} i_{i,t-1} + (1 - \rho_{i,i}) (\phi_{\pi,i} \pi_{i,t} + \phi_{y,i} \hat{y}_{i,t-1}) \quad (\text{A.3})$$

$$s_{i,t} = s_{i,t}^{\text{shock}} + 0.3 \max(-\hat{y}_{i,t-1}, 0) + 0.2 s_{i,t-1} \quad (\text{A.4})$$

$$k_{i,t} = k_{i,t}^{\text{shock}} - 2s_{i,t} + 0.5(i_{i,t} - 0.02) + 0.3 k_{i,t-1} \quad (\text{A.5})$$

$$q_{i,t} = 0.6 q_{i,t-1} + 10 g_{i,t}^e - 8(i_{i,t} + s_{i,t}) - 3 \max(-\hat{y}_{i,t}, 0) + q_{i,t}^{\text{shock}} \quad (\text{A.6})$$

$$e_{i,t} = 0.4 e_{i,t-1} + 0.5 s_{i,t} - 0.3 k_{i,t} + t_{i,t}^{\text{fx}} \quad (\text{A.7})$$

$$\Phi_{i,t}^{\text{fin}} = -0.15 s_{i,t} - 0.10 \max(-k_{i,t}, 0) \quad (\text{A.8})$$

The global oil market clears according to:

$$\Delta S_t = \lambda_t^{\text{block}} (s^H - s^{\text{Iran}}) + s^{\text{Iran}} \lambda^{\text{prod}} - 0.3 \lambda_t^{\text{block}} (s^H - s^{\text{Iran}}) \quad (\text{A.9})$$

$$\frac{\Delta P_t^{\text{oil}}}{P^{\text{oil}}} = \frac{\Delta S_t - \text{SPR}_t + \Delta D_t}{|\varepsilon_d| + \varepsilon_s} \quad (\text{A.10})$$

Financial contagion operates through:

$$\tilde{s}_{j,t}^{\text{shock}} = s_{j,t}^{\text{shock}} + 0.1 \left(\sum_{i \neq j} L_{ij} s_{i,t-1} \right) (1 + 2 \max(-k_{\text{Iran},t}, 0)) \quad (\text{A.11})$$